

With Compliments from:

5) Parallel $P = V \cdot I$

$$20\Omega : 125V \times 6.25A = 781.25W$$

$$50\Omega : 125V \times 2.5A = 312.5W$$

$$100\Omega : 125V \times 1.25A = 156.25W$$

Series $P = VI$

$$20\Omega : 25V \times 1.25A = 31.25W$$

$$30\Omega : 37.5V \times 1.25A = 46.875W$$

$$50\Omega : 62.5V \times 1.25A = 78.125W$$

LUVIS

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For T. & G. Facial Board, D/Frame Mahogany & Cyprus /

MDF Boards, Locks Handles Chrome Pipes, etc.

With Compliments from:

Embedded systems and 10T

1.4) a) Resistors in series:

$$R_T = R_1 + R_2 + R_3$$

$$R_T = 20 + 30 + 50$$

$$R_T = 100 \Omega$$

b) Resistors in parallel

$$\frac{1}{R_T} = \frac{1}{20} + \frac{1}{100} + \frac{1}{50}$$

$$= \frac{5 + 2 + 1}{100}$$

$$\frac{1}{R_T} = \frac{8}{100}$$

$$R_T = \frac{100}{8} = 12.5 \Omega$$

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2) a) ~~Parallel~~ Series

$$V = IR \therefore I = \frac{V}{R}$$

$$\frac{125}{100} = 1.25 \text{ A}$$

b) Parallel

$$I = \frac{V}{R}$$

$$\frac{125}{12.5} = 10 \text{ A}$$

3) b) Parallel

$$\text{i) } 20 \Omega \quad \frac{125}{20} = 6.25 \text{ A}$$

$$\text{ii) } 50 \Omega \quad \frac{125}{50} = 2.5 \text{ A}$$

$$\text{iii) } 100 \Omega \quad \frac{125}{100} = 1.25 \text{ A}$$

a) Series

~~For~~ In series the current through each resistor is equal to the current through the circuit:
 $\therefore I = 1.25 \text{ A}$

4) Voltage drop

Parallel circuit voltage drop across each resistor is equal to source voltage

$$V_{\text{total}} = V_1 = V_2 = V_3$$

$$= 125 \text{ V}$$

Series

$$\text{a) } 20 \Omega: 1.25 \text{ A} \times 20 \Omega = 25 \text{ V}$$

$$30 \Omega: 1.25 \text{ A} \times 30 \Omega = 37.5 \text{ V}$$

$$50 \Omega: 1.25 \text{ A} \times 50 \Omega = 62.5 \text{ V}$$